# Jekaterina Novikova

Email Scholar GitHub LinkedIn Website

## WORK EXPERIENCE

**Lead Research Scientist** | Lumina Platforms, Canada — Nov 2024 - present

- Set up a research agenda and conducted research within the area of multimodal NLP.

**Science Lead** | AI Risk and Vulnerability Alliance (non-profit), USA remote — March 2023 - present

- Co-organized the ACL SIGSEC, bridging academic and industry expertise in NLP and LLM Security.
- Presented work on inclusivity, fairness, and trust in NLP systems at multiple venues.
- Mentored a team of ML researchers and research engineers, guiding them to develop a tool for detecting and quantifying biases in NLP datasets.

**Director of ML Research** | Winterlight Labs (acquired by Cambridge Cognition), Canada — May 2018 - Apr 2024

- Set up a research agenda within the area of NLP in health.
- Managed and mentored a team of ML interns, researchers and research engineers, who published and presented research at INTERSPEECH, EMNLP, NAACL, AAAI and NeurIPS.
- Collaborated and fostered successful partnerships with research teams at top pharma companies.

**Postdoctoral Research Fellow** | Heriot-Watt University, UK — Nov 2015 - Apr 2018

- Conducted research in the fields of NLG and dialogue systems, developed the new E2E dataset and its eval.
- Published as the first author and presented at EMNLP, NAACL, SigDial (best paper nomination) and ACL.
- Co-organized the E2E shared challenge; also the INLG conference.
- Advisors: Dr. Verena Rieser, Dr. Oliver Lemon

**Visiting Researcher**| National Institute of Science, Japan — Oct 2014 - Dec 2014

- Conducted advanced research in the field of human-robot interaction, published and presented at HAI (best paper nomination).
- Advisor: Dr. Tetsunari Inamura

## EDUCATION

**PhD in Computer Science** | University of Bath, UK — 2012 - 2015
Advisors: Dr. Joanna Bryson, Dr. Leon Watts

**MSc in Computer Science** | Blekinge Institute of Technology, Sweden — 2009 - 2011

**MSc in Business Informatics** | Vytautas Magnus University, Lithuania — 2001 - 2004

**BSc in Informatics** | Vytautas Magnus University, Lithuania — 1996 - 2001

## KEY SKILLS

**Languages**: Python, LaTeX | **Frameworks**: PyTorch, HuggingFace, einops, scikit-learn, numpy/scipy, pandas | **Core technical skills**: classical machine & deep learning, NLP, LLMs, speech processing.

## AWARDS AND NOMINATIONS

- Top 50 Most Extraordinary Women Advancing AI In 2024, by GlobalBiz Outlook.
- Top 25 Women in AI in Canada, by Re-Work. 2022
- Industry Icon Award. Applied Research in Action, by the MScAC programme at the University of Toronto, 2021

## SELECT PUBLICATIONS

**Highly Cited Papers:**

- 400+ authors incl. **Jekaterina Novikova** (2023). Beyond the Imitation Game: Quantifying and extrapolating the capabilities of language models, in *Journal of Transactions on Machine Learning Research.* [total 1200+ citations as of January 2025, Featured Certification by TMLR 2024, to be presented at ICLR 2025]
- 300+ authors incl. **Jekaterina Novikova** (2022). BLOOM: A 176B-Parameter Open-Access Multilingual Language Model. arXiv:2211.05100 [total 1700+ citations as of January 2025]
- Aparna Balagopalan, Benjamin Eyre, Frank Rudzicz, **Jekaterina Novikova** (2020). To BERT or Not To BERT: Comparing Speech and Language-based Approaches for Alzheimer's Disease Detection. In *INTERSPEECH*, 2020. [total 160+ citations as of January 2025]
- **Jekaterina Novikova**, Ondrej Dusek, Amanda Cercas-Curry and Verena Rieser (2017): Why We Need New Evaluation Metrics for NLG. In *EMNLP 2017.* Copenhagen, Denmark [total 550+ citations as of January 2025]

**Impactful research:**

- 50+ authors incl. **Jekaterina Novikova** (2025). INCLUDE: Evaluating Multilingual Language Understanding with Regional Knowledge, in *ICLR 2025*.
- Aparna Balagopalan, Benjamin Eyre, Jessica Robin, Frank Rudzicz, **Jekaterina Novikova** (2021). Comparing Pre-trained and Feature-based Models for Prediction of Alzheimer's Disease Based on Speech, in *Frontiers in Aging Neuroscience*, Vol. 13, p. 189 [Impact factor: 4.8]
- Ondřej Dušek, **Jekaterina Novikova**, Verena Rieser (2019). Evaluating the State-of-the-Art of End-to-End Natural Language Generation: The E2E NLG Challenge, in *Journal of Computer Speech and Language*, Vol. 59, pp. 123-156 [Impact factor: 4.3]

## SELECT INVITED TALKS

- *Natural Language Processing for Science (NLP4Science)* workshop*,* EMNLP 2024. Panel speaker.
- *Machine Learning for Cognitive and Mental Health (ML4CMH)* workshop at AAAI 2024. Panel speaker.
- *LG Toronto AI Lab.* The Changing Face of NLP Evaluation: from Early Conversational Systems to Modern LLMs, April 2024
- *Toronto ML Summit.* Paving the Road to Trustworthy AI: Bias in Large Language Models. June 2023
- *TMLS Summit on NLP.* Interpretability and Robustness of Transformer Models in Healthcare. September 2022

## PROFESSIONAL SERVICE

**Expert Advisor**: Special Interest Group SIGSEC on NLP&LLM Security at ACL.

**Organizer**: Workshops LLMSEC@ACL'25, HEAL@CHI'25, ML4CMH@AAAI'24; Robust Evaluation Working Group at the BigScience, 2021; E2E NLG Shared Task 2017; INLG 2016.

**Chair**: Area Chair of the Workshop GEM: Natural Language Generation, Evaluation, and Metrics, EMNLP 2023, Publication Chair of the Workshop GEMv2: Natural Language Generation, Evaluation, and Metrics, EMNLP 2022.

**Steering committee**: Toronto Machine Learning Society (2020 - current).

**Mentor**: mentoring early researchers at ML Collective, Interstellar Initiative (2020-2023).

**Program committee/reviewer**: COLM (2024-2025), ACL (2020-2023), EMNLP (2017-2023), NeurIPS (2022-2023), ICLR 2021, COLING 2018, INLG (2016-2017), HRI 2016, CHI 2015, HAI (2016 - 2020), TAROS (2013-2015).

## REFERENCES

Available upon request.